Combatting Lassa Fever: Investigating Pathogenesis and Constructing a Preliminary Vaccine Candidate

INTRODUCTION

Lassa Virus (LASV)

- Lassa Fever (LF): hemorrhagic fever caused by Lassa virus (LASV) with a case fatality rate ranging from 21% to 69% and 100,000 - 300,000 cases annually (World Health Organization, 2018).
  - One of WHO's priority pathogens in need of therapeutic research

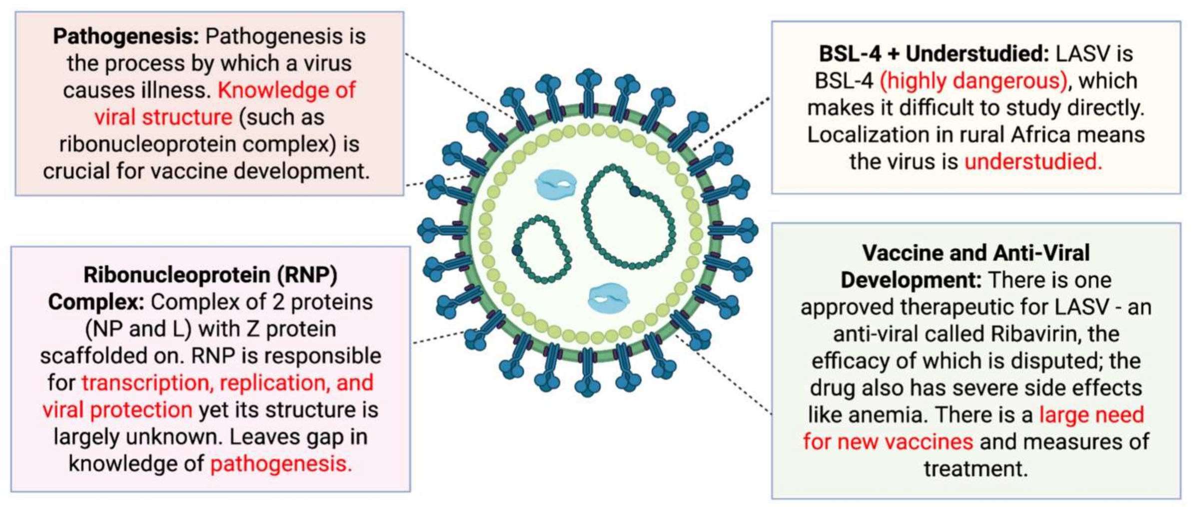


Figure 1: Difficulties in studying LASV. Made by student researcher with BioRender (2026).

Computational Methods + Synthetic Peptides

- *In Silico* RNP Complex Modelling: Predict structure from genome sequences alone (Sanger et al., 2023).
  - Molecular dynamics (MD) simulations to test movement
- Synthetic peptides: Synthesize individual epitopes from *in silico* predictions chemically instead of biologically.

Immune System Response

- T cells: white blood cells essential for adaptive immunity
  - CD8+ (cytotoxic): killer cells that destroy infected cells
  - CD4+ (helper): release signals to activate other immune cells
- B cells: crucial to humoral immunity by producing antibodies

OBJECTIVES

- This study tackles two key problems:
  - First, we look at viral structure via the RNP
  - Second, we identify LASV vaccine components (epitopes)

Goals

I.	II.	III.	IV	V.	VI.
Elucidate LASV RNP structure	Analyze structure through MD simula - tion	Build computer pipeline for rapid epitope identification	Evaluate peptide ability to induce T cell response	Evaluate peptide ability to induce humoral response	Investigate immune response mechan- ism of action

METHODOLOGY OVERVIEW

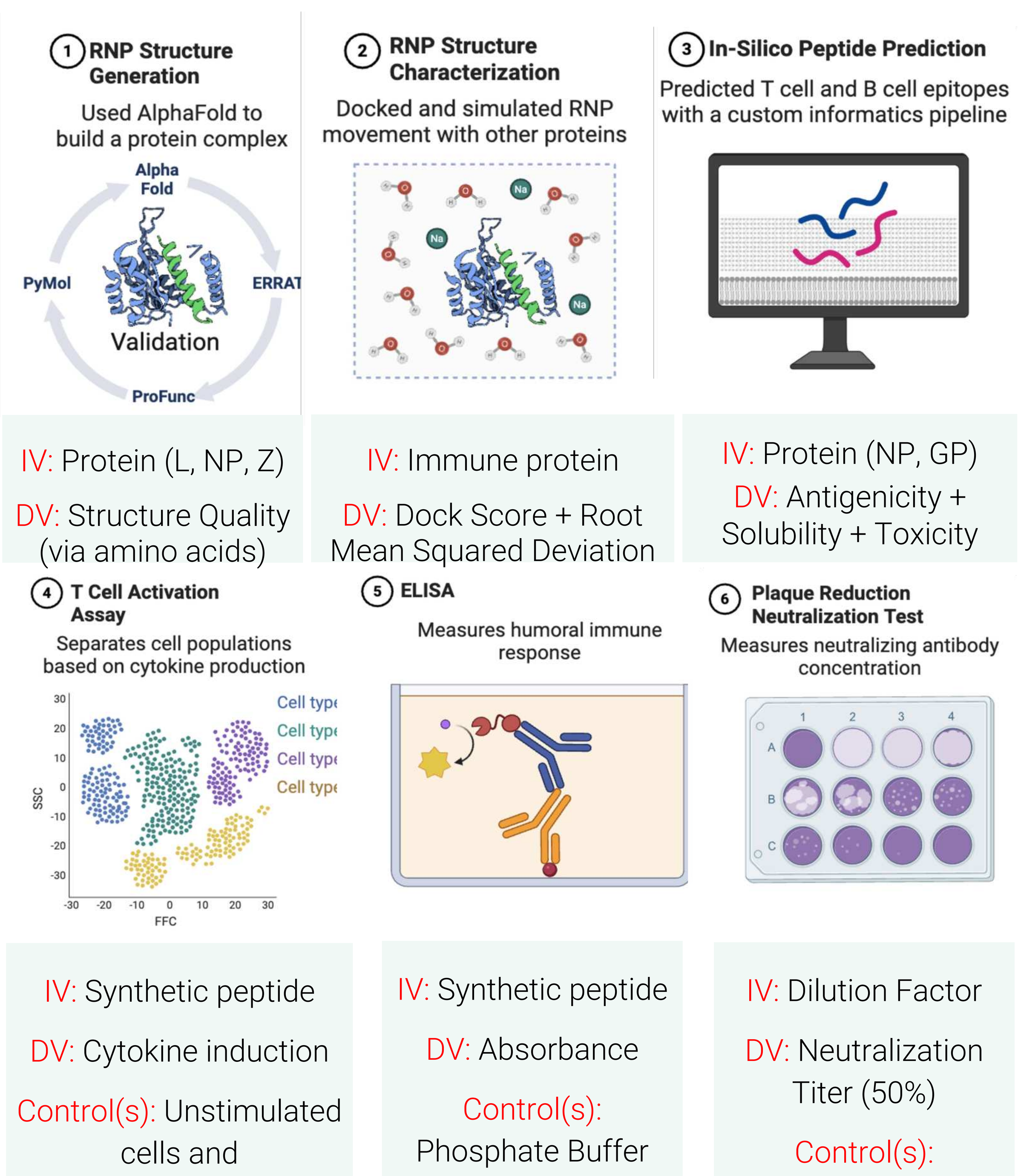


Figure 2: Methodology and variables. Made by student researcher with BioRender (2026).

I + II. METHODOLOGY

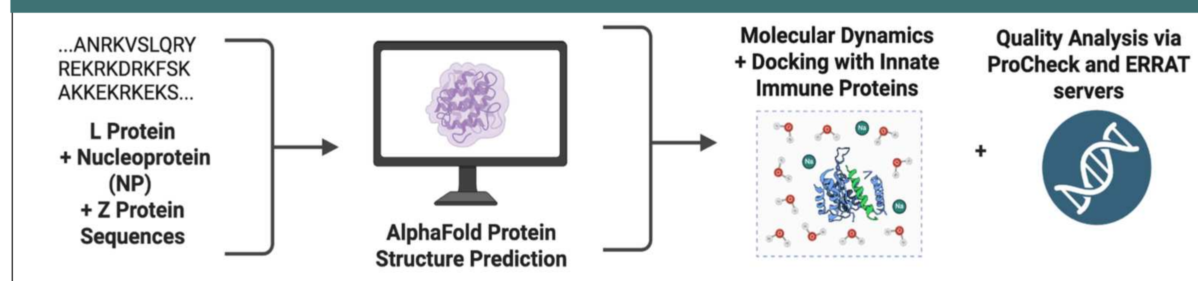


Figure 3: RNP generation/characterization methodology.

I + II. RESULTS

RNP Structure is Good Quality

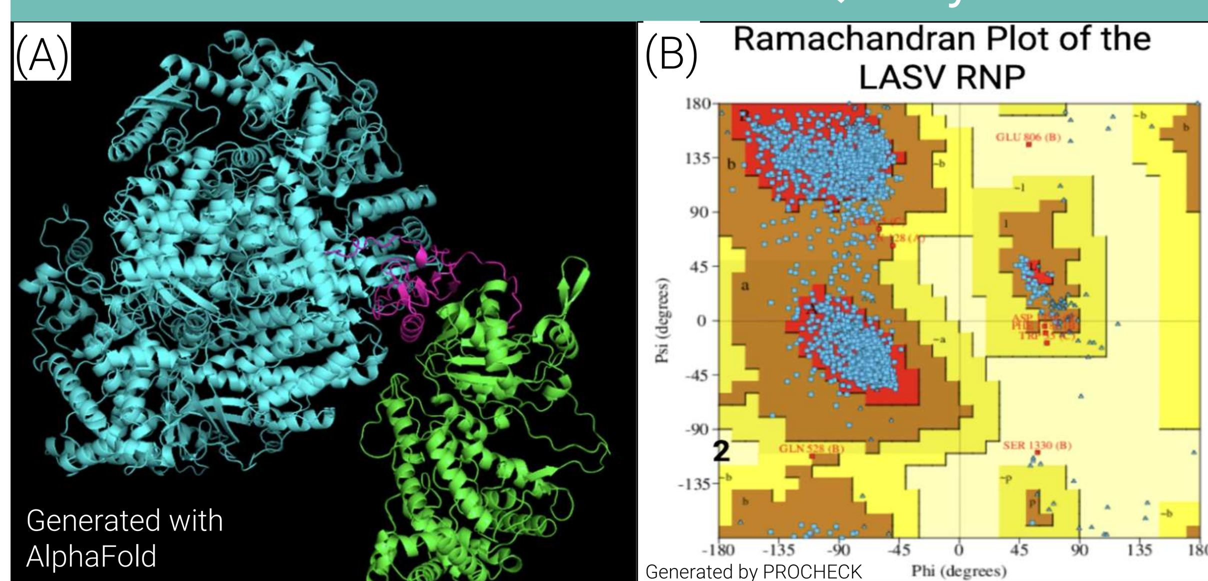
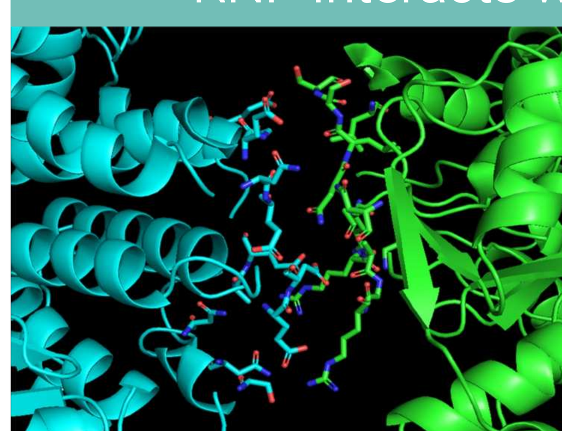


Figure 4: (A) PyMol visualization of the constructed RNP complex (blue = L protein; pink = Z protein; green = NP protein). (B) Ramachandran plot of Phi (degrees) vs Psi (degrees) of the RNP complex (91.7% of residues in the most favored regions) (2026).

- 3 protein complex successfully generated
- Protein template modelling score (PTM) = 0.65 (>0.5), indicating biological plausibility
- Ramachandran score > 90% indicates good quality

RNP Interacts with Immune Proteins



The RNP was found to have interactions with STAT-1, RIG-I, MAVS, MDA5, IKK3, and PML, human immune proteins known to be associated with LASV pathogenesis.

These binding sites could be targets of inactivation.

Figure 5: STAT-1 (blue) and RNP (green) interface. Visualized in PyMOL (2026).

III. METHODOLOGY

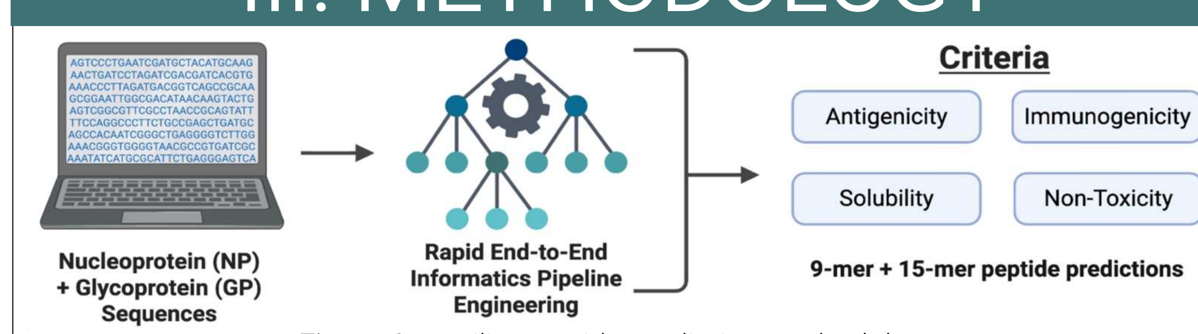


Figure 6: *In silico* peptide prediction methodology.

III. RESULTS

55 *In Silico* T-Cell and B-Cell Epitopes

Top Predicted T-Cell Activating Epitope Candidates						
Protein	Type	Peptide Sequence	Antigenicity	Solubility	Toxicity	*In Vitro* Alignment
GP	CD8+	NETGLELTL	1.1	1.60	-1.23	GP1-2
GP	CD8+	TRDIYISRR	1.72	1.88	-1.22	GP1-6
NP	CD8+	ALTDLGLIY	1.28	1.01	-1.52	NP7
NP	CD8+	ASRTSITGR	1.22	1.83	-1.09	NP11
NP	CD4+	GWPYIASRTSIT GRA	1.09	1.36	-0.69	NP10 + NP11

Table 1: Top predicted T-cell activating epitopes. Created by the researcher (2026).

Top Predicted Antibody Inducing Epitope Candidates			
Protein	Peptide Sequence	Antigenicity	*In Vitro* Alignment
GP	RTRDIYISRRLLGTF	2.06	GP1-6
GP	SKYWYLNHTTTGRTS	2.06	GP2-4
NP	GVVVEKKKRGGKEEI	1.01	NP17 + NP18
NP	GMSGGNQGARAGRD	1.03	NP5

Table 2: Top predicted Antibody inducing epitopes. Created by the researcher (2026).

55 such epitopes were identified as antigenic, soluble, and non-toxic and synthesized *in vitro*.

IV. METHODOLOGY

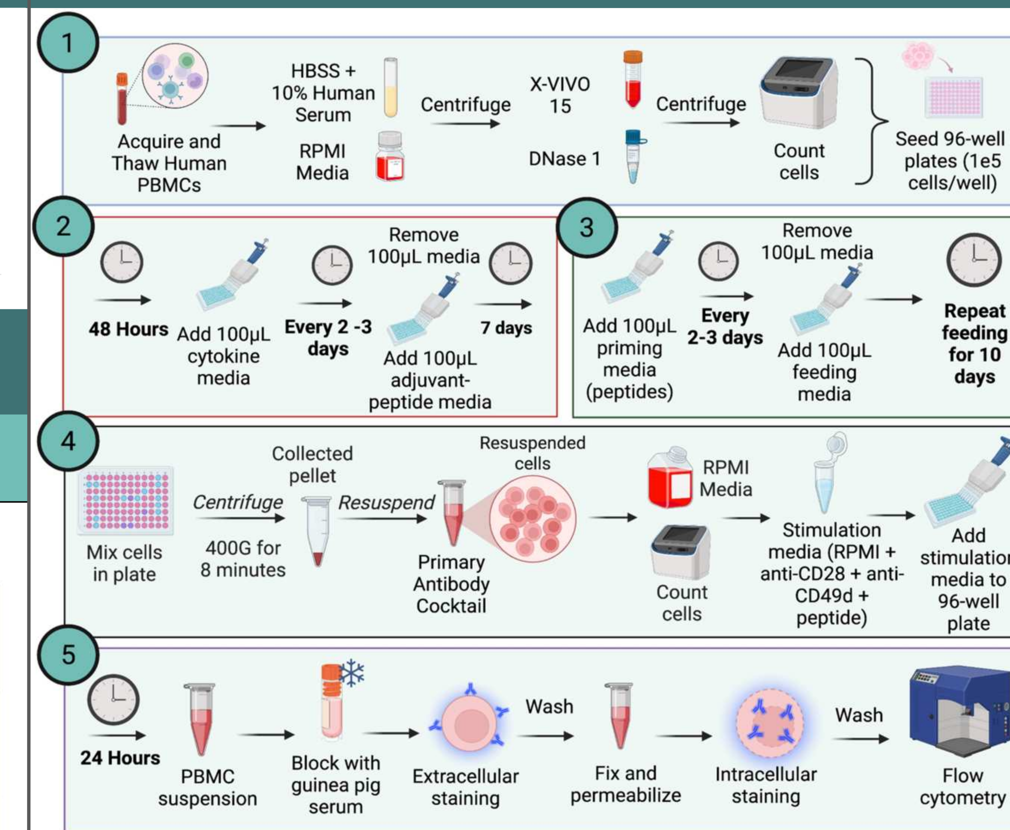
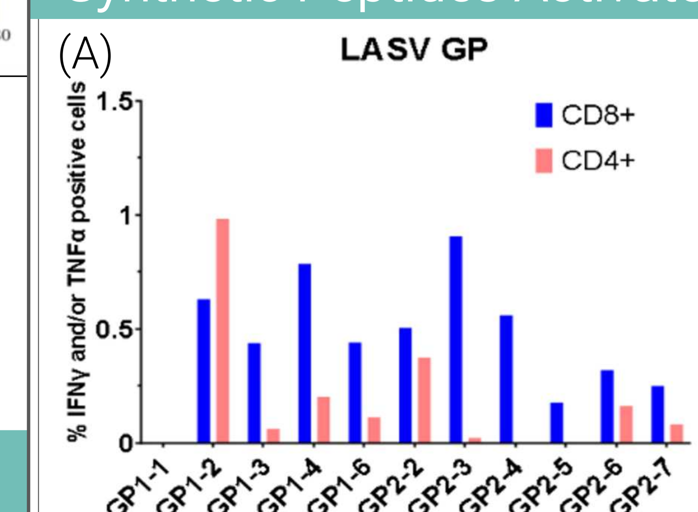
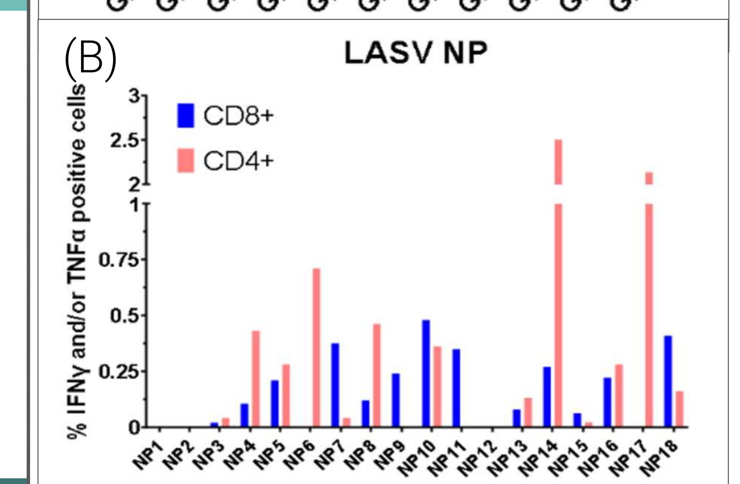


Figure 7: T-cell activation assay methods. (1) = plating (PBMCs = peripheral blood mononuclear cells), (2) = differentiation, (3) = priming, (4) = stimulation, (5) = flow cytometry. Created using BioRender (2026).

IV. RESULTS

Synthetic Peptides Activate CD8+ & CD4+ T Cells





- Predicted GP CTL epitopes overlapped with GP1-2, GP1-4, GP1-6, GP2-3, and GP2-5, all of which were CD8+ activating.
- HTL epitopes GP1-2, GP1-4, and GP2-2 activated CD4+ T-cells

- The NP CTL epitopes displayed a similar trend with NP7 and NP11.
- The HTL epitopes overlapped with NP4, NP11, and NP13.
- NP4 and NP13 were CD4+ activating.

Figure 8: CD8+ and CD4+ cell activation for each GP (A) and NP (B) peptide. The y-axis represents the percent of cells secreting either TNF-a and/or IFN-y.

V. METHODOLOGY

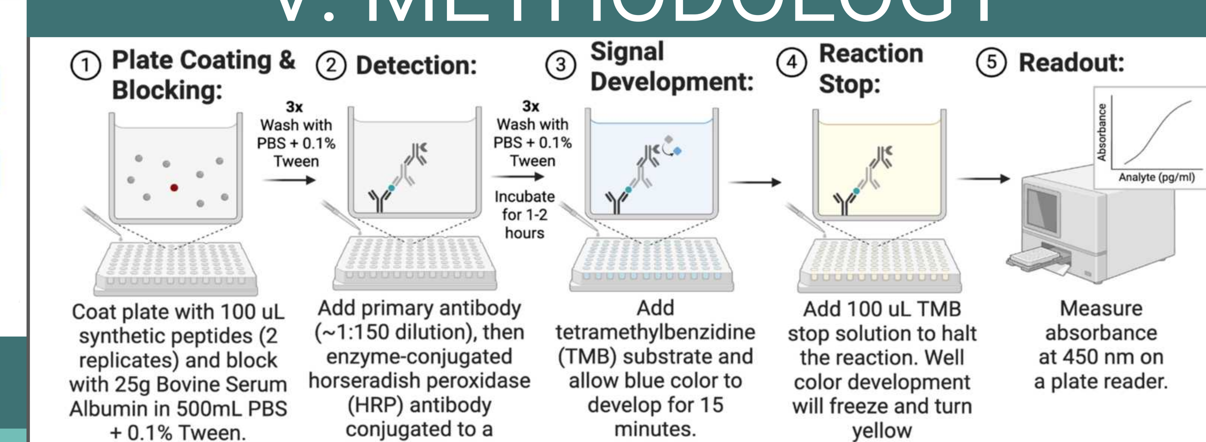
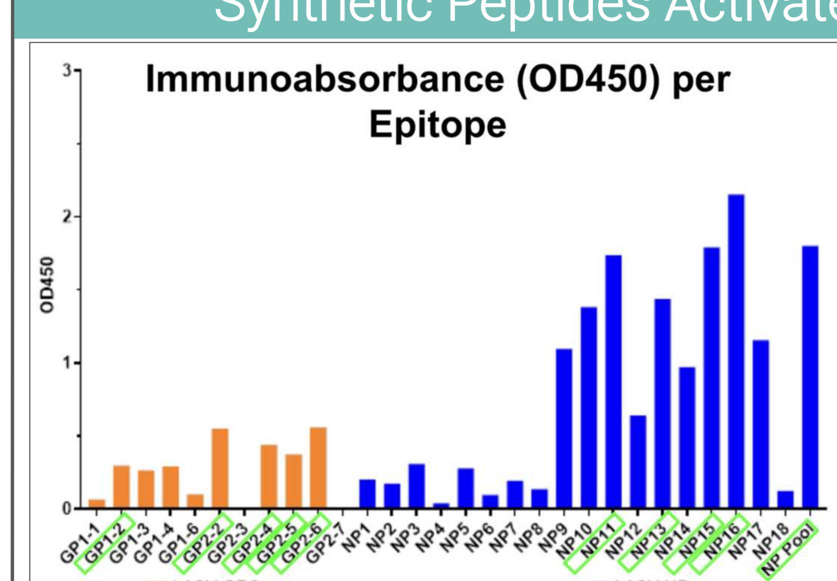


Figure 9: Humoral immunity ELISA procedure. Created by researcher in BioRender (2026).

V. RESULTS

Synthetic Peptides Activate B Cells



Between all experiments, it was found that GP1-2 induced both antibody production and CD8+ response. NP11 and GP2-6 induced both CD8+ and CD4+ activation. NP14 induced both CD4+ activation and antibody production. GP2-2 and NP10 produced all three responses.

Figure 10: OD450 (immunoabsorbance at the wavelength of 450 nm) for *in silico* identified GPC and NP peptides. Created by researcher in GraphPad (2026).

VI. METHODOLOGY

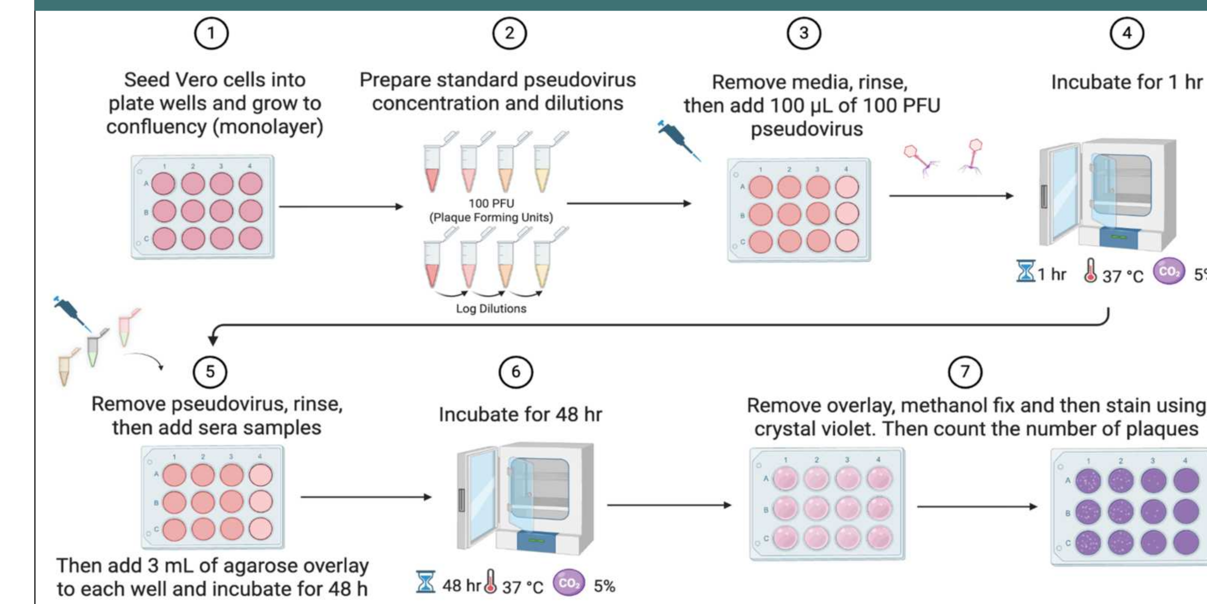
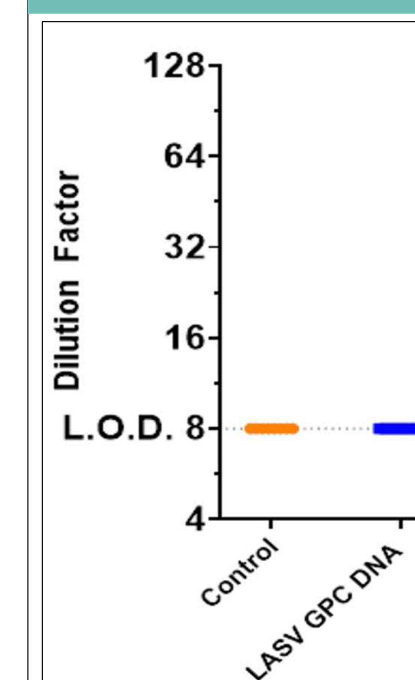


Figure 11: Plaque reduction assay methodology. Created using BioRender (2026).

VI. RESULTS

Non-Neutralizing Antibodies and T Cells MoA



No neutralizing antibodies were present in LASV GPC DNA vaccinated sera despite full protection from lethal LASV challenge (Log rank p=0.04)

This is unexpected as neutralizing antibody production is usually necessary for vaccine induced protection.

This suggests that the primary mechanism of immune response is non-neutralizing antibody production and a strong T-cell response, which provides novel insight into effective vaccine candidate designs.

Figure 12: Dilution factor at which 50% of the pseudovirion is neutralized. Created by researcher using GraphPad (2026).

CONCLUSIONS

Implications

- Identifies novel preliminary peptide vaccine candidates: peptides show T-cell and humoral immune response - could serve as parts of a vaccine with more testing.
- 1st full structure of the LASV RNP: provides a novel template to investigate pathogenesis
- Novel immune protein interactions: potential therapeutic targets.
- Immunoinformatics pipeline unlocks rapid vaccine construction capabilities: could be used for any virus/disease, drastically reducing development cost ($ billions) and time (decades)

Limitations

- The experiments were performed with synthetic peptides and a pseudovirion, which mimic but are not the same as live LASV.
- RNP structure and interaction predictions are strictly computational and don't necessarily translate to real life infection dynamics.

FUTURE WORK

1. **Test on more diverse cell populations**
   All of the cells in this study came from one human donor (1,000,000 cells per well). Studying peptides on a larger *in vitro* group could fortify results.
2. **Multi-peptide in vitro study**
   Peptides may have antagonistic or synergistic effects if applied together. Thus, a rational next step is to test combinations of synthetic peptides on the same cell population.
3. ***In vitro* nucleocapsid**
   Performing a co-immunoprecipitation experiment could validate/ modify the in silico discovered RNP structure *in vitro.*
4. ***In vivo* testing**
   Testing these peptides in a mouse or guinea pig model would help determine their real world applicability in a vaccine.

Figure 13: Future work (numbered 1-4). Created by researcher using Biorender (2026).

KEY REFERENCES

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